Physics-Informed Battery State Estimation Under Temperature & Aging

Candidate: Independent Graduate Applicant **Target Disciplines:** Applied Physics | Energy Systems | Controls **GitHub:** github.com/warlord80/battery-bms-temperature-project

> *"I independently developed a reproducible computational study of lithium-ion battery state-of-charge (SOC) estimation under coupled thermal and degradation scenarios. Using a 1-RC Thévenin equivalent-circuit model coupled to a lumped single-node thermal ODE, I compared open-loop Coulomb counting against a closed-loop Extended Kalman Filter (EKF) across 36 automated scenarios and 75 multi-seed Monte Carlo robustness runs. The study showed that while Coulomb counting accumulates severe drift (RMSE up to 15.01%) under capacity fade and sensor bias, the EKF observer maintains RMSE below 0.89% across all operating conditions and converges within 5 seconds under −15% initial state error. Tornado sensitivity analysis revealed that initial state uncertainty (±20%) and capacity fade (SOH = 0.80) dominate estimation drift, while elevated temperature (25 °C to 45 °C) primarily modulates internal ohmic overpotentials via Arrhenius kinetics. I am eager to extend this foundation through experimental parameter identification (HPPC/EIS) and embedded BMS hardware-in-the-loop validation in a graduate energy storage research group."*

CORE QUANTITATIVE FINDINGS

Scenario Dimension	Coulomb Counting (Baseline)	Extended Kalman Filter (EKF)	Physical Mechanism
Baseline (25 °C, SOH 1.0)	RMSE = 15.00%, Final err = −15.00%	**RMSE = 0.34%**, Final err = +0.00%	EKF eliminates −15% initial error in <5 s via voltage feedback.
Aging Impact (SOH = 0.80)	Divergence (Final err = +6.25%)	**RMSE = 0.53%**, Final err = +0.00%	Open-loop integration over-predicts charge as available capacity shrinks.
Elevated Temp (45 °C)	RMSE = 15.00% (unaltered)	**RMSE = 0.17%** (superior tracking)	Arrhenius activation reduces R_0 (42 to 26 mΩ), tightening OCV coupling.
Current Bias (+1.0%)	Drift = +0.50% SOC/hour	**< 0.05% residual error**	Observer innovation residual continuously suppresses sensor bias drift.

METHODOLOGY & TECHNICAL RIGOR

- **Physical Governing Model:** 1-RC Thévenin equivalent-circuit model coupled to a lumped single-node thermal energy balance ($C_{th}\, dT/dt = Q_{gen} - hA(T - T_{amb})$), enforcing non-negative irreversible dissipation ($Q_{gen} \geq 0$).
- **Sourced Cell Provenance:** Parameterized for a commercial Panasonic NCR18650B (3.20 Ah NCA/Graphite) cell based on peer-reviewed experimental literature (Barletta et al., *Energies* 2022; Mehta & Gupta, *RSER* 2024).
- **Observer Architecture:** Discrete EKF observer tracking $\mathbf{x} = [z, v_p]^T$ with analytical measurement Jacobians and Joseph-stabilized covariance updates ($\mathbf{P}_{k|k}$) guaranteeing numerical positive semi-definiteness.
- **Automated Testing Suite:** 9 unit tests verifying zero-current relaxation kinetics (τ_1 = 60 s), Faraday charge conservation, OCV strict monotonicity, and time-step convergence (Δt = 1.0 s vs. 0.5 s with <0.01% variance).
- **Reproducibility:** Single command (`python run_pipeline.py`) recreates all 36 scenarios, 75 robustness evaluations, and 8 figures in <90 s.

PROSPECTIVE GRADUATE RESEARCH ALIGNMENT

- **Electrochemical Modeling:** Extending equivalent-circuit models to Doyle–Fuller–Newman (DFN) and single-particle models (SPMe) for solid-phase concentration tracking.
- **Experimental Characterization:** Applying environmental chamber cycling, HPPC, and electrochemical impedance spectroscopy (EIS) to parameterize degradation kinetics.
- **Embedded BMS Control:** Developing real-time state-of-health (SOH), state-of-power (SOP), and thermal safety estimators for electric vehicle battery packs.

Full code, data card, test logs, and 12-page report: **github.com/warlord80/battery-bms-temperature-project** October 2026